

Average HbA1c of a Non-Insulin-Treated (NIT) Type 2 Diabetes Patient

Compiled 2026-06-07 · clinical trials + real-world registries + CGM/market sources · figures cross-checked against PubMed

NHANES

UK NDA

Swedish NDR

German DPV

IMMEDIATE

Dexcom CONNECT

SUSTAIN / AWARD / EMPA-REG

The headline — two correct answers, pick by context

~7.2%

≈ 55 mmol/mol · range 7.0–7.6%

True population average

All NIT T2D patients — mostly oral-agent, reasonably controlled

~8.6–8.8%

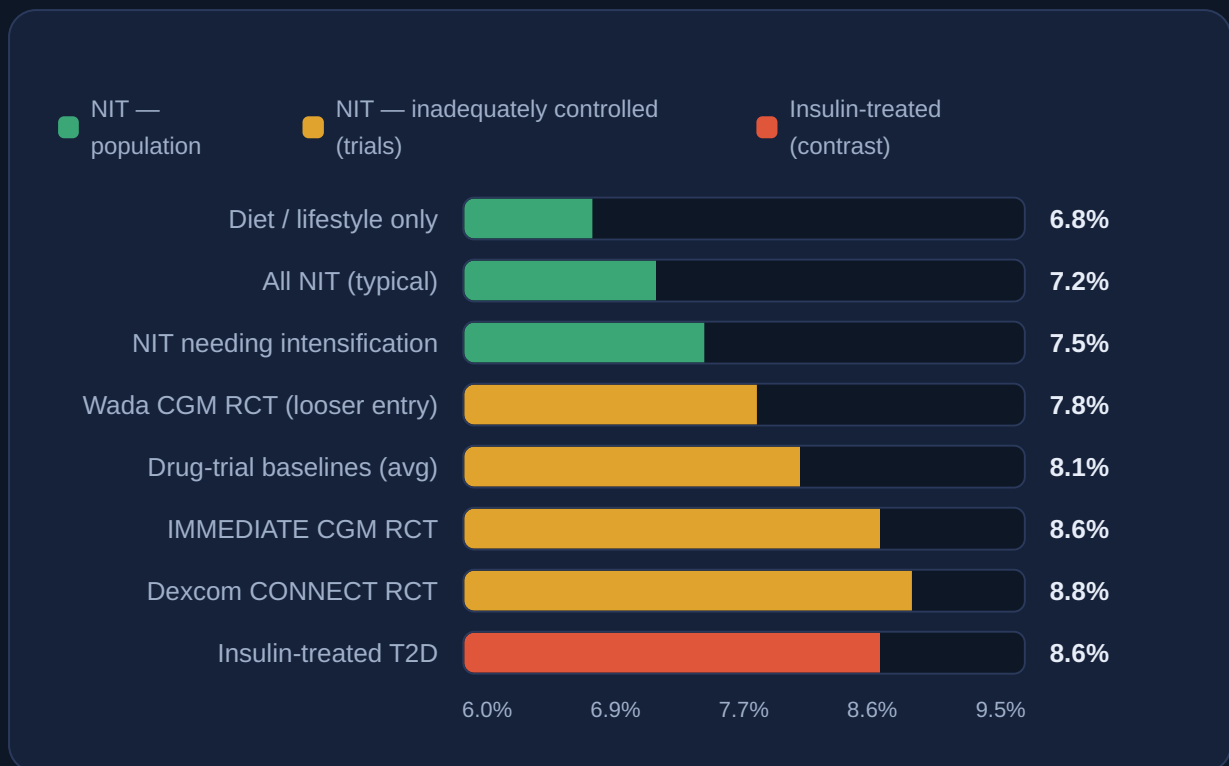
≈ 70–73 mmol/mol

Inadequately-controlled NIT

CGM / intervention-trial target population (selected above an A1c floor)

Why the gap? Pure **selection bias**. Trials enrol patients above an A1c threshold (commonly $\geq 7.5\%$), so their baselines overstate the true population mean. Never quote the 8.6% trial baseline as the population average — state which population you mean.

Where each population sits on the A1c scale



The treatment gradient — % reaching A1c target

The lighter the therapy, the better the control — diet > oral > insulin. (Partly **confounded by indication**: insulin is started in longer / more severe disease, so this is not a causal effect of therapy.)

NHANES III — % achieving HbA1c <7%



Population target-attainment anchors

SOURCE	METRIC	RESULT
NHANES 1999–2004	T2D <7%	52.2%
NHANES — metformin monotherapy	<7%	62.2%
UK National Diabetes Audit 24/25	≤7.5% (≤58 mmol/mol)	64.7%
NHANES insulin users (2013–20)	<7%	27.5%

Source evidence

Real-world registries & surveys

POPULATION

SOURCE	A1C / METRIC
Swedish NDR (whole T2D)	~6.7–7.0%
German DPV/DIVE (median)	~7.1–7.2%
NHANES 1999–2004	52% <7%
UK NDA 24/25	65% ≤7.5%
CPRD/MASTERMIND (intensify)	>7.5%

Synthesis → ~7.2% (55 mmol/mol)

CGM / intervention RCTs TRIAL

BASELINE

TRIAL	BASELINE	N
IMMEDIATE 2023	8.6%	116
Dexcom CONNECT 2026	8.8%	283
Martens 2025	~8.5%	72
Vigersky 2012	~8.7%	100
Wada 2020	7.8%	100

Synthesis → 8.6–8.8% (70–73 mmol/mol)

Pharma RCT baselines — oral / GLP-1 / SGLT2i / DPP-4 add-on TRIAL BASELINE

TRIAL	BACKGROUND	BASELINE A1C	N
AWARD-3 (2014)	monotherapy / drug-naive	~7.6%	807
EMPA-REG MET (2014)	metformin only	7.9%	637
DPP-4 + metformin (2006)	metformin	~8.0%	701
SUSTAIN 2 (2017)	metformin / TZD	8.1%	1,225
SUSTAIN 8 (2019)	metformin only	8.3%	788
LEAD-3 (liraglutide)	monotherapy	8.3%	~746

Add-on-for-inadequate-control cluster 7.9–8.3%; monotherapy runs lower (~7.6%).

⚠ **Common miscitation:** **MOBILE** (JAMA 2021, baseline 9.1%) and the **DIAMOND T2D** cohort are **basal/MDI-insulin-treated** INSULIN — **not** NIT. Exclude them from NIT figures despite their frequent appearance in "CGM in T2D" searches.

Recommended phrasing

HONEST POPULATION STATEMENT

"The average non-insulin-treated T2D patient has an HbA1c of roughly 7.2% (55 mmol/mol), versus ~8.6% (70 mmol/mol) for insulin-treated patients."

DEVICE / MARKET DECK

"Inadequately-controlled non-insulin T2D — the CGM target population — has a baseline A1c of ~8.6–8.8% (IMMEDIATE, CONNECT); CGM lowers A1c by ~0.3–0.9% vs control."

Caveats — read before quoting any number

- ⚠ **Two populations, two numbers** — never quote the 8.6% trial baseline as the population average.
- ⚠ **Selection bias** inflates every trial baseline (entry floors of A1c ≥ 7.1 – 7.5%).
- ⚠ **"Non-insulin" is heterogeneous** — diet-only (~ 6.5 – 7.0%) < metformin-only < multi-oral/GLP-1.
- ⚠ **Confounding by indication** drives the NIT-vs-insulin gap (disease stage, not therapy).
- ⚠ **MOBILE / DIAMOND-T2D are insulin-treated** — exclude from NIT figures.
- ⚠ **CONNECT is press-sourced** (ADA 2026); reconfirm exact figures vs the peer-reviewed manuscript.
- ⚠ **No clean NIT-specific A1c distribution** (NHANES-style %<7/7–8/8–9/>9) was verifiable; the population mean is a reasoned synthesis, not a single published figure.

Key references

1. Aronson R et al. **IMMEDIATE** — isCGM in non-insulin T2D. *Diabetes Obes Metab* 2023;25(4): 1024–31. DOI (PMID 36546594)
2. Wada E et al. Flash GM vs SMBG, non-insulin T2D. *BMJ Open Diab Res Care* 2020;8(1):e001115. DOI
3. Martens TW et al. CGM to guide food choices, non-insulin T2D. *Diabetes Technol Ther* 2025;27:261–70. DOI
4. Dodd AH et al. Treatment approach & A1c control, NHANES 1999–2004. *Curr Med Res Opin* 2009. DOI
5. Hankosky ER et al. A1c targets, insulin-treated, NHANES 2009–2020. *Diabetes Ther* 2023. DOI
6. Venkatraman S et al. Glycemic control in insulin users, NHANES 1988–2020. *JAMA Netw Open* 2022. DOI
7. Whyte MB et al. Glycaemic control disparities, T2D England (CPRD). *PLoS Med* 2019. DOI
8. Rawshani A et al. Newly-diagnosed T2D, Swedish NDR. *BMJ Open* 2015. DOI
9. Dexcom CONNECT (ADA 2026) — Dexcom IR; ADA
10. UK National Diabetes Audit — digital.nhs.uk
11. ADA Standards of Care 2025 — diabetes.org

Peer-reviewed figures cross-checked against PubMed; press-sourced figures (CONNECT) flagged. Unit conversion: $\text{mmol/mol} = (\% - 2.15) \times 10.929$ (NGSP/DCCT $\leftrightarrow$ IFCC). Bar-chart positions are illustrative mappings of the cited point estimates onto a 6.0–9.5% scale.